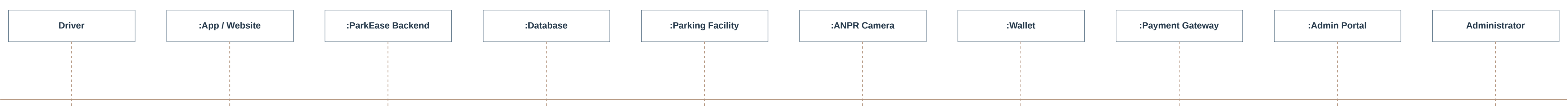
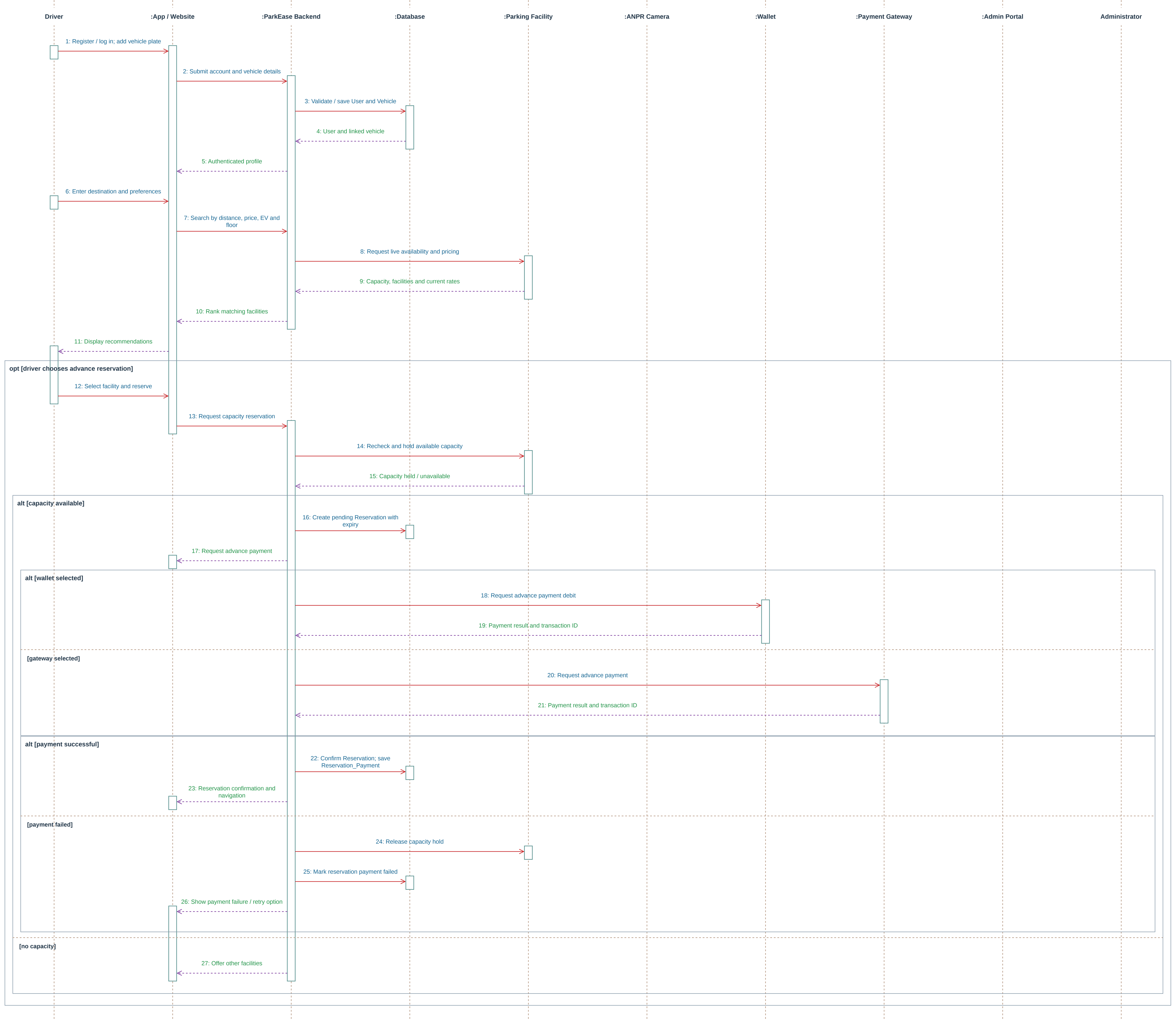
ParkEase - Complete Sequence Diagram

One-page model: booking, automatic entry, billing, timed exit and independent admin activities.


Driver
:App / Website
:ParkEase Backend
:Database
:Parking Facility
:ANPR Camera
:Wallet
:Payment Gateway
:Admin Portal
Administrator

01 - Search and reservation

Optional advance reservation holds capacity. Continue to entry after successful booking or as an eligible walk-in.


Driver
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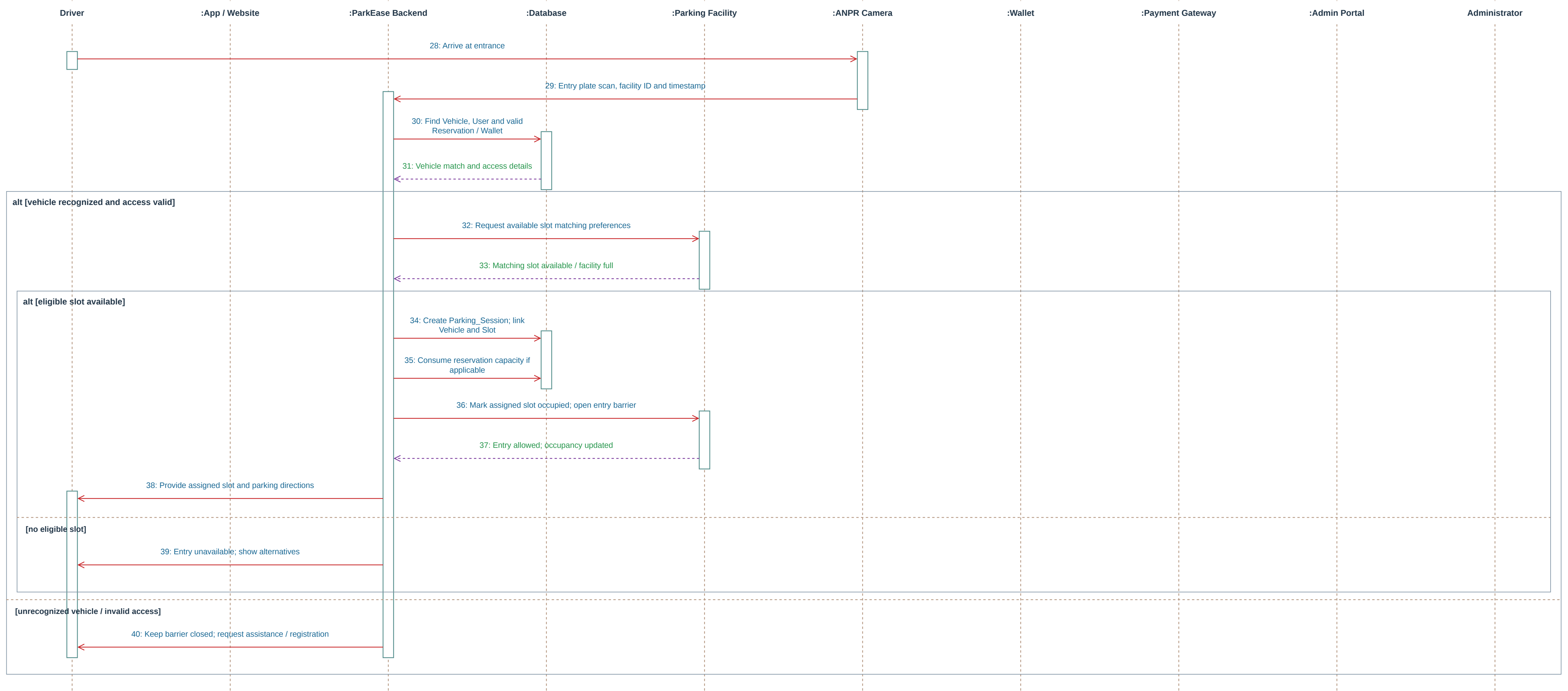
Administrator
1: Register / log in; add vehicle plate
2: Submit account and vehicle details
3: Validate / save User and Vehicle
4: User and linked vehicle
5: Authenticated profile
6: Enter destination and preferences
7: Search by distance, price, EV and floor
8: Request live availability and pricing
9: Capacity, facilities and current rates
10: Rank matching facilities
11: Display recommendations
opt [driver chooses advance reservation]
12: Select facility and reserve
13: Request capacity reservation
14: Recheck and hold available capacity
15: Capacity held / unavailable
alt [capacity available]
16: Create pending Reservation with expiry
17: Request advance payment
alt [wallet selected]
18: Request advance payment debit
19: Payment result and transaction ID
[gateway selected]
20: Request advance payment
21: Payment result and transaction ID
alt [payment successful]
22: Confirm Reservation; save Reservation_Payment
23: Reservation confirmation and navigation
[payment failed]
24: Release capacity hold
25: Mark reservation payment failed
26: Show payment failure / retry option
[no capacity]
27: Offer other facilities

Continue to automatic entry only after successful booking, or use the eligible linked-vehicle / wallet flow without a reservation.

02 - Automatic entry

Continue with a valid reservation or linked vehicle / wallet. Verify the plate before allocating the exact slot.


Driver
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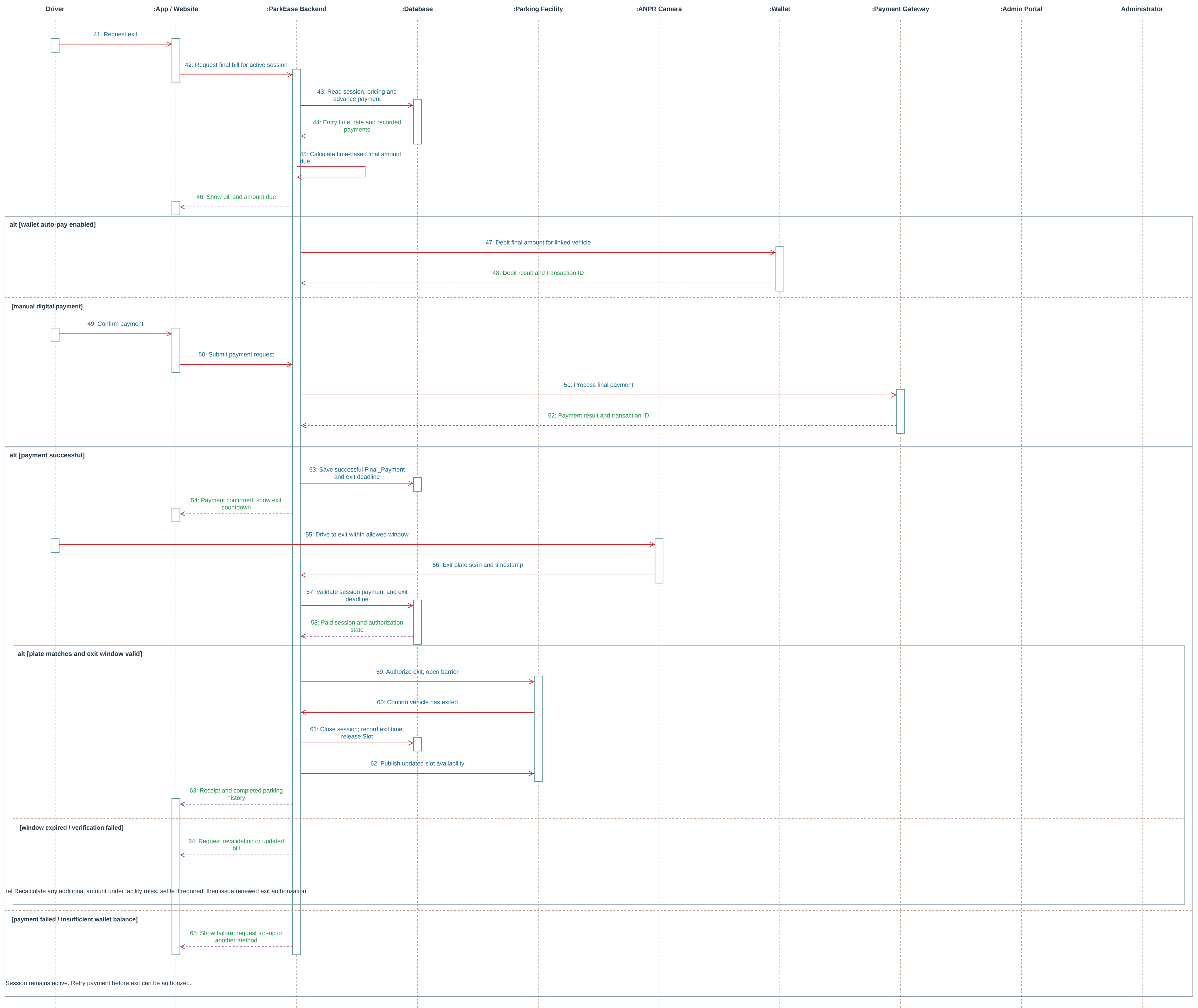
28: Arrive at entrance
29: Entry plate scan, facility ID and timestamp
30: Find Vehicle, User and valid Reservation / Wallet
31: Vehicle match and access details
alt [vehicle recognized and access valid]
32: Request available slot matching preferences
33: Matching slot available / facility full
alt [eligible slot available]
34: Create Parking_Session; link Vehicle and Slot
35: Consume reservation capacity if applicable
36: Mark assigned slot occupied; open entry barrier
37: Entry allowed; occupancy updated
38: Provide assigned slot and parking directions
[no eligible slot]
39: Entry unavailable; show alternatives
[unrecognized vehicle / invalid access]
40: Keep barrier closed; request assistance / registration

The active session records entry time and the slot assigned at arrival. Continue to billing when the driver requests exit.

03 - Billing and exit

Continue with an active parking session. Successful payment grants a 10-minute exit window plus configurable grace.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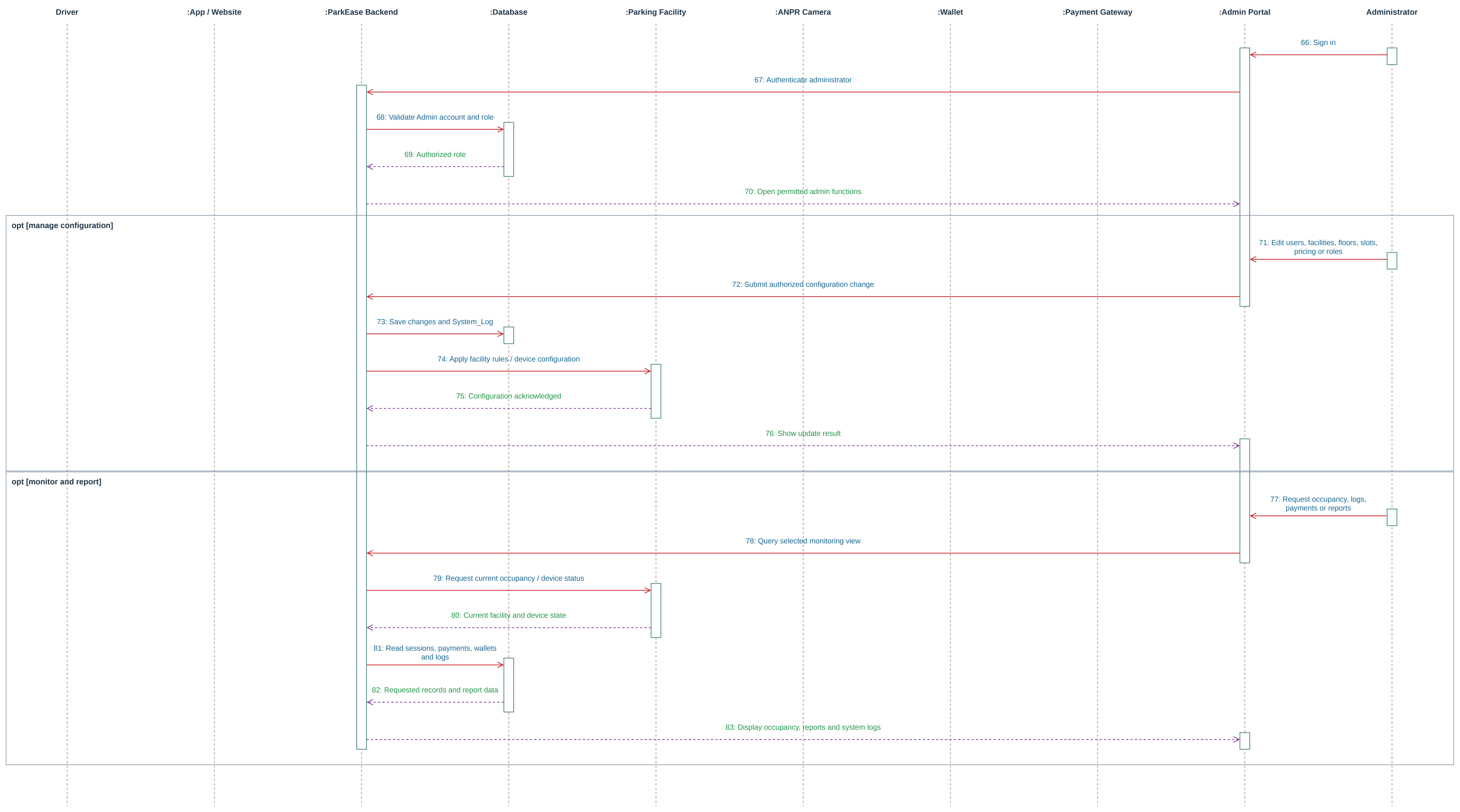
Administrator
41: Request exit
42: Request final bill for active session
43: Read session, pricing and advance payment
44: Entry time, rate and recorded payments
45: Calculate time-based final amount due
46: Show bill and amount due
alt [wallet auto-pay enabled]
47: Debit final amount for linked vehicle
48: Debit result and transaction ID
[manual digital payment]
49: Confirm payment
50: Submit payment request
51: Process final payment
52: Payment result and transaction ID
alt [payment successful]
53: Save successful Final_Payment and exit deadline
54: Payment confirmed; show exit countdown
55: Drive to exit within allowed window
56: Exit plate scan and timestamp
57: Validate session payment and exit deadline
58: Paid session and authorization state
alt [plate matches and exit window valid]
59: Authorize exit; open barrier
60: Confirm vehicle has exited
61: Close session; record exit time; release Slot
62: Publish updated slot availability
63: Receipt and completed parking history
[window expired / verification failed]
64: Request revalidation or updated bill
ref Recalculate any additional amount under facility rules, settle if required, then issue renewed exit authorization.
[payment failed / insufficient wallet balance]
65: Show failure; request top-up or another method
Session remains active. Retry payment before exit can be authorized.

Billing detail: apply any eligible advance-payment credit using facility rules; do not charge the same amount twice.

04 - Admin management

Independent administrative interactions; these are not mandatory steps in the driver journey.


Driver
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:Admin Portal
Administrator
66: Sign in
67: Authenticate administrator
68: Validate Admin account and role
69: Authorized role
70: Open permitted admin functions
opt [manage configuration]
71: Edit users, facilities, floors, slots, pricing or roles
72: Submit authorized configuration change
73: Save changes and System_Log
74: Apply facility rules / device configuration
75: Configuration acknowledged
76: Show update result
opt [monitor and report]
77: Request occupancy, logs, payments or reports
78: Query selected monitoring view
79: Request current occupancy / device status
80: Current facility and device state
81: Read sessions, payments, wallets and logs
82: Requested records and report data
83: Display occupancy, reports and system logs

Red arrows / blue labels: requests | Purple dashed arrows / green labels: returns | Narrow bars: activations | opt: optional | alt: alternatives | ref: linked interaction